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Can LLMs Uncover Hidden Physical Laws through Free Exploration?

LongChang Benchmarks Open-Ended Physical Discovery in Altered-Physics MuJoCo Worlds: 11 of 12 Frontier Models Score Zero

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Eleven of twelve frontier large language models fail entirely at open-ended physical discovery: placed in a MuJoCo simulation with silently altered physical laws and given tools to run experiments freely, they detect anomalies but cannot formulate the governing equation. Only GPT-6-Astra succeeds at all, with a 16.7% discovery rate. The result documents a sharp gap between competent reasoning within known frameworks and genuine hypothesis-driven scientific inquiry.

AT A GLANCE

Problem: It is unknown whether LLMs can perform genuine open-ended scientific discovery—designing experiments, iterating hypotheses, and formulating laws without being prompted.

Why it matters: Autonomous AI scientists could accelerate research, but only if their discovery capabilities go beyond pattern matching on known science.

Method: LongChang framework: MuJoCo worlds with silently altered physics; LLM agents given Python tools for free exploration; discovery scored by law-formulation accuracy.

Result: 11 of 12 frontier models score 0%; GPT-6-Astra achieves 16.7% (1 of 6 trials).

Evidence: 12 frontier LLMs evaluated across 6 altered-physics environments; qualitative analysis of exploration patterns.

The Big Picture

Scientific discovery requires more than retrieval or deduction from known facts. It requires *abduction*: noticing that the world behaves unexpectedly, designing experiments to distinguish candidate explanations, accumulating evidence, and arriving at a formal law. This process is epistemically distinct from the question-answering and reasoning tasks on which LLMs currently excel.

The question addressed by this paper is sharply defined: given a simulation world in which one or more physical constants have been silently altered, and given a set of Python tools with which to run experiments, can an LLM figure out which law has changed and state it correctly? No hint is given about which quantity to vary. No labeled data is provided. The agent must decide what to measure, how to measure it, and what to conclude.

Why Existing Approaches Fall Short

Equation discovery benchmarks (e.g., SRBench, AI Feynman) provide labeled (x_i, y_i) pairs and ask models to fit a symbolic expression. This is regression, not discovery: the model is told which variables are relevant and is given cleaned, ready-made data.

Scientific reasoning benchmarks (e.g., SciBench) test whether LLMs can answer physics questions correctly. This requires application of known laws, not their formulation.

Agent-with-tool benchmarks (e.g., WebArena, SciAgent) give agents access to search or computation tools but define the task question explicitly. The agent does not decide what question to ask.

LongChang differs on all three dimensions: no labeled data, no known applicable law, no predefined question.

Core Mechanism

Altered-Physics Worlds. A MuJoCo simulation is configured with one or more modified constants: gravitational acceleration g, coefficient of restitution e, or air drag coefficient k. The alteration is sampled from a set of plausible values (e.g., $g \in \{4, 6, 14, 18\}\,\mathrm{m/s^2}$) to prevent trivial detection.

Tool Suite. The agent is provided with Python tools to:

- spawn and reset objects (balls, ramps, pendulums),
- apply forces or initial velocities,
- query sensor readings (position, velocity, force) over time,
- log numerical results to a file.

The agent writes and executes Python code calling these tools.

Exploration Protocol. The agent receives a budget of N tool calls (typically $N = 200$). It is told only that "some physical properties of this world may differ from the standard physics you know." It must determine what those properties are and state them formally by the end of its budget.

Scoring. A discovery is marked successful if the agent's final stated law is numerically correct within $\pm 5\%$ of the true altered value, and identifies the correct physical quantity. Partial credit is given for correct anomaly

identification alone.

Technical Formulation

Let $\mathcal{W}_c$ denote a world with altered physical constant $c \in \mathcal{C}$. The agent is a language model $\mathcal{A}$ that generates a sequence of tool calls $a_1, a_2, \ldots, a_N$ and a terminal law statement $\hat{L}$.

The discovery score is:

$$S(\mathcal{A}, \mathcal{W}_c) = \mathbf{1}[|\hat{c} - c|/c < \epsilon \ \wedge \ \hat{q} = q]$$

where $\hat{c}$ is the agent's estimated value, q is the correct quantity, and $\epsilon = 0.05$.

The Collective Anomaly Detection score (CAD) measures partial credit:

$$\text{CAD}(\mathcal{A}) = \mathbb{E}_{\mathcal{W}_c}[\mathbf{1}[\text{agent flags anomaly in quantity } q]]$$

The gap between CAD and S quantifies the failure mode: detecting that something is wrong but not determining what.

Learning or Inference Procedure

No training is performed; all models are evaluated zero-shot with their standard system prompts. The evaluation is therefore a direct test of models' out-of-the-box capability for autonomous scientific discovery. Each model is given a standardized initial message describing the tools and the open-ended task.

Chain-of-thought prompting is used (the model is asked to reason before each tool call), but no few-shot examples of discovery are provided to avoid biasing the exploration strategy.

What the Guarantee Says

No formal guarantee is claimed. The paper's contribution is empirical: it establishes a lower bound of ≈0% on the discovery ability of current frontier models (with one exception), and provides a repeatable benchmark for tracking progress. The LongChang framework's automatic world generation ensures that future evaluations can use fresh worlds not seen in any training data, preventing benchmark saturation.

Experimental Findings

- **Discovery rates:** GPT-6-Astra: **16.7%**; all other 11 models: **0%**.
- **Anomaly detection (CAD):** GPT-6-Astra: 66.7%; average across all models: 41.2%. Models can detect that something is wrong far more reliably than they can determine what is wrong.
- **Hypothesis revision:** All models converge on an initial hypothesis after the first 20–50 tool calls and rarely revise it, even when subsequent measurements contradict it.
- **Tool use quality:** Models tend to run many single-variable experiments but fail to design the controlled comparisons needed to isolate the altered constant.
- **Gravity vs. drag:** Altered gravity is discovered more often than altered drag (GPT-6-Astra discovers gravity in all successful trials), consistent with gravity's more direct effect on commonly simulated trajectories.

Ablations and Interpretation

Provided anomaly hint: When the agent is told "the gravitational acceleration in this world has been altered," success rates rise to 50%+ for GPT-6-Astra and 16.7% for several other models. The bottleneck is therefore anomaly targeting, not law formulation.

Reduced tool budget ($N = 50$): GPT-6-Astra's discovery rate drops from 16.7% to 0%; it needs ∼150 calls to triangulate the correct value. Other models are equally unsuccessful at $N = 50$.

Single-alteration vs. multi-alteration worlds: Success rates drop to 0% for all models when two constants are simultaneously altered, suggesting that current LLMs cannot disentangle compound perturbations.

Failure mode categorization: The authors classify failures as: (1) no anomaly detected (17%), (2) anomaly detected but wrong quantity identified (51%), and (3) correct quantity but incorrect value (32%). Type (3) is encouraging for the future: the bottleneck is precise estimation, not total blindness.

Reference

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